

- Stefan A. Frisch, Nathan R. Large, and David B. Pisoni. Perception of wordlikeness: Effects of segment probability and length on the processing of nonwords. *Journal of memory and language*, 42:481–496, 2000. URL <https://api.semanticscholar.org/CorpusID:1736965>.
- Leonidas Gee, Andrea Zugarini, Leonardo Rigutini, and Paolo Torroni. Fast vocabulary transfer for language model compression. In Yunyao Li and Angeliki Lazaridou (eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pp. 409–416, Abu Dhabi, UAE, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-industry.41. URL <https://aclanthology.org/2022.emnlp-industry.41/>.
- Martin Gerlach and Francesc Font-Clos. A standardized project gutenber corpus for statistical analysis of natural language and quantitative linguistics, 2018. URL <https://arxiv.org/abs/1812.08092>.
- Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih (eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 5484–5495, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.446. URL <https://aclanthology.org/2021.emnlp-main.446>.
- Mor Geva, Avi Caciularu, Kevin Wang, and Yoav Goldberg. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang (eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 30–45, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.3. URL <https://aclanthology.org/2022.emnlp-main.3>.
- Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. Dissecting recall of factual associations in auto-regressive language models. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 12216–12235, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.751. URL <https://aclanthology.org/2023.emnlp-main.751>.
- Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A unifying framework for inspecting hidden representations of language models. In *Proc. of ICML*, volume abs/2401.06102, 2024. URL <https://arxiv.org/abs/2401.06102>.
- Mandy Guo, Zihang Dai, Denny Vrandečić, and Rami Al-Rfou. Wiki-40B: Multilingual language model dataset. In Nicoletta Calzolari, Frédéric Béchet, Philippe Blache, Khalid Choukri, Christopher Cieri, Thierry Declerck, Sara Goggi, Hitoshi Isahara, Bente Maegaard, Joseph Mariani, Hélène Mazo, Asuncion Moreno, Jan Odijk, and Stelios Piperidis (eds.), *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pp. 2440–2452, Marseille, France, May 2020. European Language Resources Association. ISBN 979-10-95546-34-4. URL <https://aclanthology.org/2020.lrec-1.297/>.
- Wes Gurnee, Neel Nanda, Matthew Pauly, Katherine Harvey, Dmitrii Troitskii, and Dimitris Bertsimas. Finding neurons in a haystack: Case studies with sparse probing. *Transactions on Machine Learning Research*, 2023. ISSN 2835-8856. URL <https://openreview.net/forum?id=JYs1R9IMJr>.
- Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A. Smith. Don’t stop pretraining: Adapt language models to domains and tasks. In Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (eds.), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 8342–8360, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.740. URL <https://aclanthology.org/2020.acl-main.740/>.

- Valentin Hofmann, Hinrich Schuetze, and Janet Pierrehumbert. An embarrassingly simple method to mitigate undesirable properties of pretrained language model tokenizers. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 385–393, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-short.43. URL <https://aclanthology.org/2022.acl-short.43>.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. Mistral 7b, 2023. URL <https://arxiv.org/abs/2310.06825>.
- Ayush Kaushal and Kyle Mahowald. What do tokens know about their characters and how do they know it? In Marine Carpuat, Marie-Catherine de Marneffe, and Ivan Vladimir Meza Ruiz (eds.), *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 2487–2507, Seattle, United States, July 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.naacl-main.179. URL <https://aclanthology.org/2022.naacl-main.179>.
- Zixuan Ke, Yijia Shao, Haowei Lin, Tatsuya Konishi, Gyuhak Kim, and Bing Liu. Continual pre-training of language models. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=m_GDIIItaI3o.
- Seungduk Kim, Seungtaek Choi, and Myeongho Jeong. Efficient and effective vocabulary expansion towards multilingual large language models. *arXiv preprint arXiv:2402.14714*, 2024.
- Stav Klein and Reut Tsarfaty. Getting the## life out of living: How adequate are word-pieces for modelling complex morphology? In *Proceedings of the 17th SIGMORPHON workshop on computational research in phonetics, phonology, and morphology*, pp. 204–209, 2020.
- Taku Kudo. Subword regularization: Improving neural network translation models with multiple subword candidates. In Iryna Gurevych and Yusuke Miyao (eds.), *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 66–75, Melbourne, Australia, July 2018. Association for Computational Linguistics. doi: 10.18653/v1/P18-1007. URL <https://aclanthology.org/P18-1007>.
- Vedang Lad, Wes Gurnee, and Max Tegmark. The remarkable robustness of llms: Stages of inference?, 2024. URL <https://arxiv.org/abs/2406.19384>.
- William Marslen-Wilson, Lorraine K Tyler, Rachelle Waksler, and Lianne Older. Morphology and meaning in the english mental lexicon. *Psychological review*, 101(1):3, 1994.
- Fanwang Meng, Michael Richer, Alireza Tehrani, Jonathan La, Taewon David Kim, Paul W Ayers, and Farnaz Heidar-Zadeh. Procrustes: A python library to find transformations that maximize the similarity between matrices. *Computer Physics Communications*, 276:108334, 2022a.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. *Advances in Neural Information Processing Systems*, 35:17359–17372, 2022b.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations*, 2017. URL <https://openreview.net/forum?id=Byj72udxe>.
- Jack Merullo, Carsten Eickhoff, and Ellie Pavlick. Language models implement simple Word2Vec-style vector arithmetic. In Kevin Duh, Helena Gomez, and Steven Bethard (eds.), *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 5030–5047, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.281. URL <https://aclanthology.org/2024.naacl-long.281>.
- nostalgebraist. interpreting GPT: the logit lens. *LessWrong*, 2020. URL <https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens>.

- Koyena Pal, Jiuding Sun, Andrew Yuan, Byron Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state. In Jing Jiang, David Reitter, and Shumin Deng (eds.), *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*, pp. 548–560, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.conll-1.37. URL <https://aclanthology.org/2023.conll-1.37>.
- Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The geometry of categorical and hierarchical concepts in large language models. In *ICML 2024 Workshop on Mechanistic Interpretability*, 2024. URL <https://openreview.net/forum?id=KXuYjuBzKo>.
- Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The representation geometry of features and hierarchy in large language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=bVTM2QKYuA>.
- Aleksandar Petrov, Emanuele La Malfa, Philip Torr, and Adel Bibi. Language model tokenizers introduce unfairness between languages. *Advances in Neural Information Processing Systems*, 36, 2023.
- Ivan Provilkov, Dmitrii Emelianenko, and Elena Voita. BPE-dropout: Simple and effective subword regularization. In Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault (eds.), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 1882–1892, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.170. URL <https://aclanthology.org/2020.acl-main.170>.
- Kathleen Rastle, Jonathan Harrington, and Max Coltheart. 358,534 nonwords: the arc nonword database. *The Quarterly journal of experimental psychology. A, Human experimental psychology*, 55:1339–62, 11 2002. doi: 10.1080/02724980244000099.
- Hassan Sajjad, Nadir Durrani, Fahim Dalvi, Firoj Alam, Abdul Khan, and Jia Xu. Analyzing encoded concepts in transformer language models. In Marine Carpuat, Marie-Catherine de Marnette, and Ivan Vladimir Meza Ruiz (eds.), *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 3082–3101, Seattle, United States, July 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.naacl-main.225. URL <https://aclanthology.org/2022.naacl-main.225/>.
- Craig W Schmidt, Varshini Reddy, Haoran Zhang, Alec Alameddine, Omri Uzan, Yuval Pinter, and Chris Tanner. Tokenization is more than compression. *arXiv preprint arXiv:2402.18376*, 2024.
- Peter H Schönemann. A generalized solution of the orthogonal procrustes problem. *Psychometrika*, 31(1):1–10, 1966.
- Neha Sengupta, Sunil Kumar Sahu, Bokang Jia, Satheesh Katipomu, Haonan Li, Fajri Koto, William Marshall, Gurpreet Gosal, Cynthia Liu, Zhiming Chen, Osama Mohammed Afzal, Samta Kamboj, Onkar Pandit, Rahul Pal, Lalit Pradhan, Zain Muhammad Mujahid, Massa Baali, Xudong Han, Sondas Mahmoud Bsharat, Alham Fikri Aji, Zhiqiang Shen, Zhengzhong Liu, Natalia Vassilieva, Joel Hestness, Andy Hock, Andrew Feldman, Jonathan Lee, Andrew Jackson, Hector Xuguang Ren, Preslav Nakov, Timothy Baldwin, and Eric Xing. Jais and jais-chat: Arabic-centric foundation and instruction-tuned open generative large language models, 2023. URL <https://arxiv.org/abs/2308.16149>. arXiv:2308.16149.
- Rico Sennrich, Barry Haddow, and Alexandra Birch. Neural machine translation of rare words with subword units. In Katrin Erk and Noah A. Smith (eds.), *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1715–1725, Berlin, Germany, August 2016. Association for Computational Linguistics. doi: 10.18653/v1/P16-1162. URL <https://aclanthology.org/P16-1162>.
- Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT rediscovers the classical NLP pipeline. In Anna Korhonen, David Traum, and Lluís Màrquez (eds.), *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 4593–4601, Florence, Italy, July 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1452. URL <https://aclanthology.org/P19-1452>.